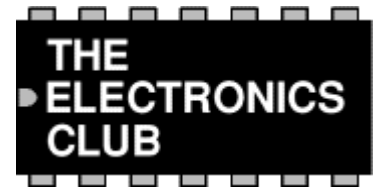


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Tools required for electronics



Also see: [Starter kit](#) | [Workbench](#) | [Soldering guide](#)

[Rapid Electronics](#) stock a wide range of tools and they have kindly allowed me to use their photographs on this page.

Soldering iron

For electronics work the best type is one powered by mains electricity (230V in the UK), it should have a **heatproof cable for safety**. The iron's power rating should be 15 to 25W and it should be fitted with a small bit of 2 to 3mm diameter.



Photograph © Rapid Electronics

Other types of soldering iron

Low voltage soldering irons are available, but their extra safety is undermined if you have a mains lead to their power supply! **Temperature controlled** irons are excellent for frequent use, but not worth the extra expense if you are a beginner. **Gas-powered** irons are designed for use where no mains supply is available and are not suitable for everyday use. Pistol shaped **solder guns** are far too powerful and cumbersome for normal electronics use.

Soldering iron stand

You must have a safe place to put the iron when you are not holding it. The stand should include a sponge which can be dampened for cleaning the tip of the iron.



Photograph © Rapid Electronics

Desoldering pump (solder sucker)



A tool for removing solder when desoldering a joint to correct a mistake or replace a component.

Photograph © Rapid Electronics

Solder remover wick (copper braid)

This is an alternative to the desoldering pump shown above.



Photograph © Rapid Electronics

Reel of solder

The best size for electronics is 22swg (swg = standard wire gauge).

Photograph © Rapid Electronics



Side cutters

For trimming component leads close to the circuit board.

Photograph © Rapid Electronics



Wire strippers

Most designs include a cutter as well, but they are not suitable for trimming component leads.

Photograph © Rapid Electronics



Small pliers

Usually called 'snipe nose' pliers, these are for bending component leads etc. If you put a strong rubber band across the handles the pliers make a convenient holder for parts such as switches while you solder the contacts.

Photograph © Rapid Electronics



Small flat-blade screwdriver

For scraping away excess flux and dirt between tracks, as well as driving screws!

Photograph © Rapid Electronics



Heat sink

You can buy a special tool, but a standard crocodile clip works just as well and is cheaper.

Photograph © Rapid Electronics



The following tool is only required if you are using stripboard:



Track cutter

A 3mm drill bit can be used instead, in fact the tool is usually just a 3mm drill bit with a proper handle fitted.

Photograph © Rapid Electronics

The following tools are only required if you make your own PCBs:

PCB rubber

This is an abrasive rubber for cleaning PCBs. It can also be used to clean stripboard where the copper tracks have become dull and tarnished.



Photograph © Rapid Electronics

Small electric drill

Ideally this should be mounted in a drill stand. You will need a range of small drill bits, but for most holes a 1mm bit is suitable. Larger holes can be drilled with a hand drill but 1mm bits are too fragile to use reliably in a hand drill.



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